IOWA STATE UNIVERSITY

Knowledge-Grounded 3D Disease Editing: What Does 3D Add Beyond 2D?

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Pilot results and three focused tests of what 3D adds
Advisor update

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The Question We Have to Answer

What does knowledge-guided 3D give us that the original 2D photo, or a knowledge-guided 2D edit, does not?

01 **Not enough: 2D → 3D → 2D**
Rendering a reconstruction back to the input view adds nothing the photo did not have.

02 **Candidate gains**
Shape symptoms (curl, pucker, galls), hidden surfaces (leaf underside), views a camera never took, one consistent object across views.

03 **Our rule**
Every claim must be measured, and the measurement must say whether it is *real*, *synthetic* or *unverified*.

How this deck is organised

Act I. Question, five hypotheses, pipeline, status.
Act II. Protocol: audit, case selection, data tiers, structured KB, matched arms.
Act III. Experiments E1–E5, three focused tests of what 3D adds, ablations, statistics, failures.
Act IV. Go / no-go, answers to the 12 questions, next steps.

Status. Pilot runs complete on all 15 specimens; three focused tests added. Numbers carry their *n* and are regenerated from the result files (2026-10-15 04:22).

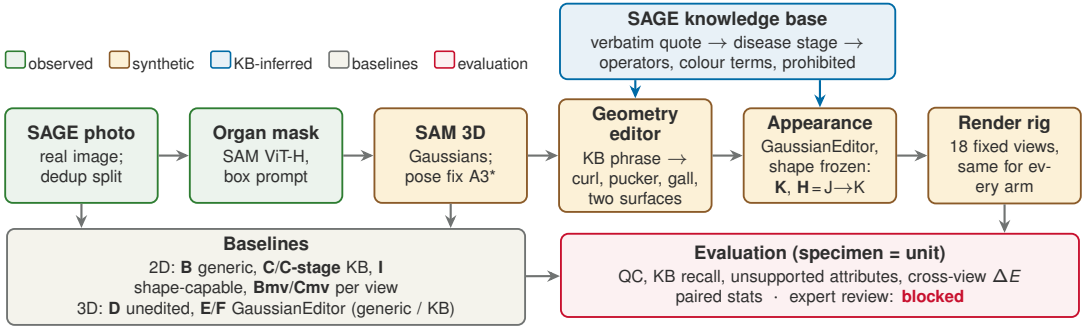
Five Hypotheses and Where Each Stands

	Hypothesis	Evidence so far (specimen = unit)	Status
H1	Single-image reconstruction preserves real anatomy	Input view: SAM 3D pose is off; a pose-only fit lifts median *fitted* IoU 0.63→0.92 (15/15). Unseen views: *no real reference*.	● input view only
H2	KB improves symptom fidelity over generic prompts	2D colour recall: KB 0.71 vs generic 0.38. 3D shape: KB operators 0.44 vs generic 0.07 (n=6). GaussianEditor: no KB shape at all.	● supported, small *n*, no expert
H3	3D editing is more multi-view consistent than 2D editing	Same-point colour difference: per-view 2D 16.8–19.3 vs 3D 9.3–10.4 (floor 2.0).	✓ internal only
H4	3D holds verifiable information missing from the input view	3D can *represent* surface, direction and volume (Tests 1–3); no suitable real views among the datasets examined; capture protocol written.	NOT RUN / BLOCKED
H5	KB 3D assets improve a downstream task	Minimal real-photo test (5 crops): no gain over unedited renders; below real-only at $k \leq 4$.	✗ not supported

Bottom line. Supported: a *controllable, auditable KB→shape editor* with consistent views. Not shown: that the 3D content is *true* (H4) or *useful* (H5).

OUR RUN – MEASURED results/alignment/summary.json, results/statistics/{comparisons.json, metrics2d_summary.json, crossview_corr_*.json}. Values regenerated 2026-10-15 04:22.

Pipeline as Implemented



Two editors. GaussianEditor changes colour, not shape (E2); shape comes from our *geometry editor* (KB terms without an operator are logged, never approximated). Three streams stay separate: observed (kept from the photo), KB-inferred (a cited phrase), synthetic (filled by a model, unverified).

OUR RUN – MEASURED diagram traced from the code paths (src/figures/fig1_pipeline.py lists them). SAM 3D Objects, GaussianEditor (Chen *et al.*, CVPR 2024), InstructPix2Pix (Brooks *et al.*, CVPR 2023).

Method: One Evidence-Conditioned Objective

$$G^* = \arg\min_G \ \lambda_{\mathrm{KB}}\mathcal{L}_{\mathrm{KB}} + \lambda_{\mathrm{anat}}\mathcal{L}_{\mathrm{anat}} + \lambda_{\mathrm{reg}}\mathcal{L}_{\mathrm{reg}} \quad \text{s.t.} \quad \mathcal{L}_{\mathrm{obs}} \le \tau$$

Term	Implemented as	Status
$\mathcal{L}_{\mathrm{KB}}$	operator targets from the cited phrase: signed curl, pucker, outgrowth, closed galls, two surfaces; evidence placement	✓ implemented
$\mathcal{L}_{\mathrm{anat}}$	as-rigid-as-possible edge lengths; galls only on kernel tissue	✓ ablation: without it strain p95 0.12/0.41 → 0.35/1.75
$\mathcal{L}_{\mathrm{reg}}$	Laplacian smoothness; non-target parts anchored	● smoothness has no measurable effect
$\mathcal{L}_{\mathrm{obs}} \le \tau$	fitted input-view IoU drop vs the real mask, hard check	✓ preserve / counterfactual / abstain

Not yet: uncertainty over geometries (counts, sizes and positions are sampled, not represented), a differentiable $\mathcal{L}_{\mathrm{obs}}$, species-specific anatomy, and a learned KB→operator map. Novelty claim: source grounding with abstention plus the photo as a constraint, not the operators.

OUR RUN – MEASURED reports/16_method_formalisation.md; results/review/b9_ablation (Mango 0049 / Peach 0087, curl up).

Three Focused Tests: What Could 3D Add?

01 Direction
Can a cited direction (“curl upwards”) be imposed and measured, with the opposite direction as a control?

02 Volume
Can a gall be a closed volume that replaces kernels, where a height field only pushes the surface out?

03 Surface
Can a lower-surface symptom exist without appearing on the upper surface, which is all a photo shows?

Design. Smallest tests that can fail: 3 specimens per disease, 3 seeds, every method rendered from the same 18 cameras and the corrected input camera, each with a placement or direction control. Original assets are never modified.

OUR RUN – MEASURED experiments/demos/run_demos.py; results/demos/execution_log.jsonl. Results: Tests 1–3 after E2.

Phase 1: Cases Where 3D Could Matter

Case	SAGE class	KB text (verbatim source)	What the photo lacks
A geometry	Peach Leaf Curl	"leaves become severely distorted (thickened and puckered)"	pucker depth, far side of curl
B occlusion	Tomato Leaf Mold	"velvety mold forms on the **lower** leaf surface"	underside never photographed
C volumetric	Corn Smut	"tumor-like galls … less than 1 cm to more than 30 cm"	gall volume, back of the ear
D relief	Citrus Canker	"corky with a crater-like depression"	relief under one light
Z control	Mango Anthracnose	flat dark spots on a near-planar leaf	nothing material

How they were chosen

Regex screen of 635 KB classes for 7 shape/occlusion families → read the source clause → inspect images → disease type from SAGE. 3 specimens per class, picked by fixed criteria *before* any model ran.

Not in SAGE or deferred

Tea tortrix, root-knot nematode: pests, absent. Grape Leafroll: canopy shots only. Black Knot: 40 images. Squash Downy Mildew: label-noise flag.

KB TEXT – VERBATIM SOURCE quotes from SAGE KB source pages (Ohioline PLPATH-FRU-26, UMN Extension, APSnet, CDFA). data/metadata/candidate_cases.csv; reports/phase1_protocol/01_case_selection.md.

Phase 2: Ground-Truth Tiers

Tier 1: real multi-view

Same specimen, several real views; one input, rest held out.

Status: none. Every SAGE image is a single, independent view with no specimen ID.

NOT RUN / BLOCKED

Tier 2: measured geometry

Scans or hand-measured traits (curl angle, gall size).

Status: none available.

NOT RUN / BLOCKED

Tier 3: public single view

SAGE: 820,502 image rows, 635 classes with source-grounded KB text.

Status: in use, 15 pilot specimens.

REAL SAGE DATA

Consequence. Without Tier 1/2, every novel view and hidden surface is *synthetic and unverified*. H1 (unseen views) and H4 cannot be tested; we measure fidelity to the input view, KB adherence and internal consistency, and label everything else.

Leakage rules in place: content-hash deduplication before any split (16,973 duplicate rows found); masks from the input image only; no held-out image enters reconstruction, prompting or editing.

REAL SAGE DATA data/metadata/sage_shard_manifest.parquet (all 48 shards); reports/phase1_protocol/02_dataset_protocol.md.

Phase 3: Structured, Stage-Aware KB

Real record: Peach Leaf Curl, stage 0 of 3

```
"source_quote":  "Developing leaves become severely
distorted (thickened and puckered), and have a reddish or
purple cast.  Later, ...  powdery gray ..."
"evidence_status":  "source_supported_unobserved"
"geometry":  [distort ← "severely distorted",
              pucker, thicken ← "thickened and puckered"]
"appearance":  [reddish, purple ← "reddish or purple cast"]
"prohibited":  [stage 1 "powdery gray",
              stage 2 "turn yellow or brown and drop",
              "no geometry operator without a cited phrase",
              "preserve background and unaffected organs"]
```

01 One stage per edit

Mixing stages (red *and* powdery grey) was a bug in the interim prompts; later stages become prohibited edits.

02 Every edit cites a phrase

Operators without a cited phrase are refused; terms without an operator (curl, roll, “corky”) are logged as unsupported.

03 Evidence level kept

Observed in the photo / source-supported but unobserved / hedged / unsupported – never merged.

04 Matched prompts

generic, joint, appearance-only, geometry-only, wrong-stage, other-disease.

KB TEXT – VERBATIM SOURCE data/kb/structured_specs.json (generated by src/kb_grounding/structured.py from the SAGE KB); source: ohioline.osu.edu/factsheet/plpath-fru-26.

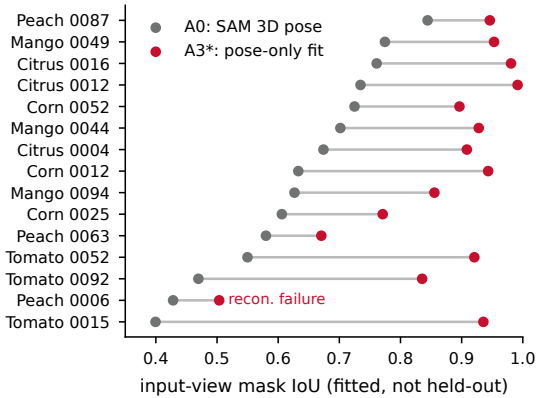
Phase 4: Matched Experimental Arms

Arm	Method	shape?	views	KB	Tests
A	real photo	–	1	–	observed information
B / C	IP2P 2D edit inside the organ mask	no	1	none / KB	is knowledge alone enough?
C-NG / C-MM	2D, KB wrong stage / other disease	no	1	control	negative controls
I	2D, dilated mask (can change outline)	yes	1	KB stage	fairest 2D shape baseline
Bmv/Cmv	IP2P on each rendered view independently	no	8	none / KB	per-view 2D consistency
D	SAM 3D, no edit	–	18	–	reconstruction alone
E / F	GaussianEditor, generic / KB prompt	in principle	18	none / KB	text-driven 3D editing
G / J	geometry editor, random field / KB operators	yes	18	none / KB	does KB control shape?
K / H	GaussianEditor, geometry frozen / J then K	K no, H yes	18	KB	colour only; full method

Fairness controls. 2D and 3D arms receive the *same* KB text. G's random field is matched to J's displacement magnitude (RMS), so only the *direction* of change differs. All 3D arms use one fixed 18-view rig and one renderer. Seeds: 3 for geometry and 2D arms; 1 so far for GaussianEditor arms (38 min each).

OUR RUN – MEASURED reports/05_baseline_comparison.md; experiments/phaseA/run_geometry.py, run_ge_arms.py; src/editing2d/ip2p_batch.py.

E1: The Error Is the Pose, Not the Camera



0.63 → 0.92

median *fitted* input-view IoU (15/15 improved)

01 Camera checked first
Intrinsics fit the depth map to <2 px (12/15); recentring changes IoU ≤0.01.

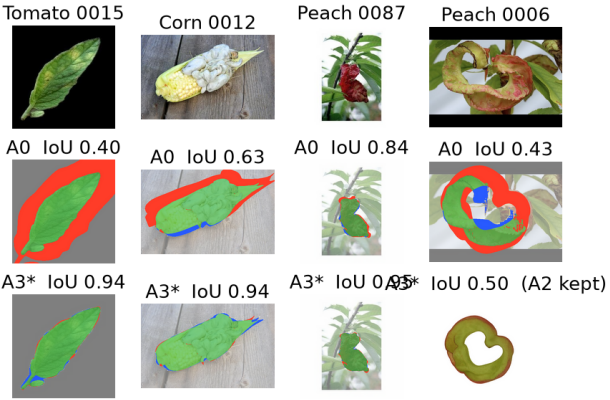
02 Objects come out too large
13/15 oversized (up to +150%); SAM 3D's pose refinement is off by default.

03 Fix touches pose only
Scale/shift, then 7-DoF render-and-compare, kept only if IoU improves (A3 alone diverged on 3/15).

04 Held-out checks are mixed
Depth agreement worse on 6/15; LPIPS worse on 1/15.

OUR RUN – MEASURED results/alignment/summary.json; reports/01_camera_alignment_fixes.md. IoU is optimised and scored on the same mask: a fit, not accuracy. 3D accuracy and held-out views: not measurable (no reference). Failure threshold 0.6 set post hoc.

E1: Examples, Including a Real Failure



What works

Leaves, fruit and ears are recovered as one object; after the pose fix the outline matches the photo closely (Tomato 0015: 0.40 $\rightarrow$ 0.94).

What fails

Peach 0006, a strongly curled leaf, becomes a **closed ring**. That is a topology error; no pose change fixes it (best IoU 0.50). Kept in every table.

What we cannot say

Whether the back or underside is right. No second real view exists.

REAL SAGE DATA photos (top row) **OUR RUN – MEASURED** overlays: green = render and mask agree, red = render only, blue = mask only. results/alignment/<specimen>/.

E2: GaussianEditor Changes Colour, Not Shape



All panels SYNTHETIC renders on the same fixed rig; nothing here is verified against a real view.

E/F: jitter, not deformation

Displacement coherence 0.31 / 0.31 (independent noise = 0.25); KB geometric recall 0.

Texture cost

Grain $\times$13.6 (E), $\times$17.0 (F) vs unedited; leaf about twice as thick.

Geometry editor (G, J)

Coherence 0.97 / 0.92; grain $\times$1.1 / $\times$1.2. J follows the KB.

SYNTHETIC – UNVERIFIED Peach Leaf Curl 0087, all arms, same 18-view rig. OUR RUN – MEASURED medians over specimens: E n=15, F n=15, G n=9, J n=9. Prune-aware Gaussian matching: results/phaseA/*/<arm>/seed*/gaussian_diff.json.

E2: KB-Driven Shape Edits on Gaussians

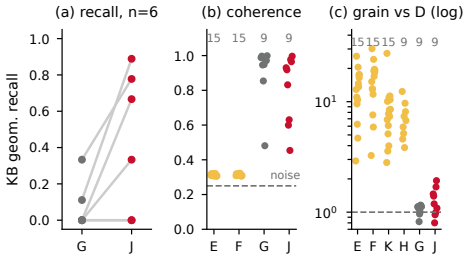
Method

$$\mu_i' = \mu_i + m_i\, d_i$$

$$\mathcal{L} = \mathcal{L}_{\text{edit}} + 0.1\,\mathcal{L}_{\text{ARAP}} + 0.01\,\mathcal{L}_{\text{smooth}} + \mathcal{L}_{\text{anchor}}$$

KB term → operator → target field h on the surface (8-NN graph). Rotations follow the new normals.

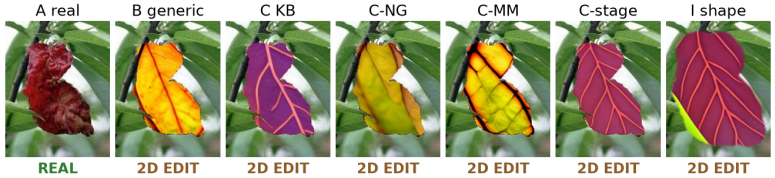
KB term	operator
puckered	Poisson-disk blisters
distorted	low-frequency warp
thickened	thicker along own normal
galls, tumors	outgrowths
sunken	depressions
curl, roll (with direction)	v2 signed curl (Test 1)
corky	✗ none yet (logged)

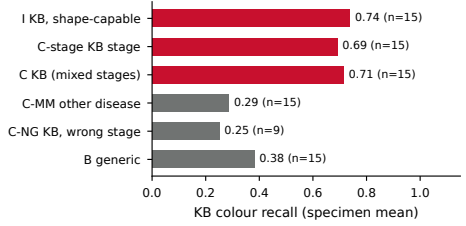


Three separate levels. (1) *Operator compliance* (here): KB vs generic, equal magnitude, recall 0.44 vs 0.07 (+0.37, CI +0.13 to +0.61; 2 classes), partly by construction. (2) *Biological fidelity*: not measured (ratings pending). (3) *Practical validity*: not shown (E5).

OUR RUN – MEASURED results/statistics/comparisons.json, specimen_metrics.csv; src/geometry_editor/deform.py. Detector thresholds set on a synthetic sheet; blister count → density change is **POST-HOC CHANGE (LOGGED)**.

E2: Knowledge Steers Colour in 2D Too





Knowledge, not 3D, drives colour

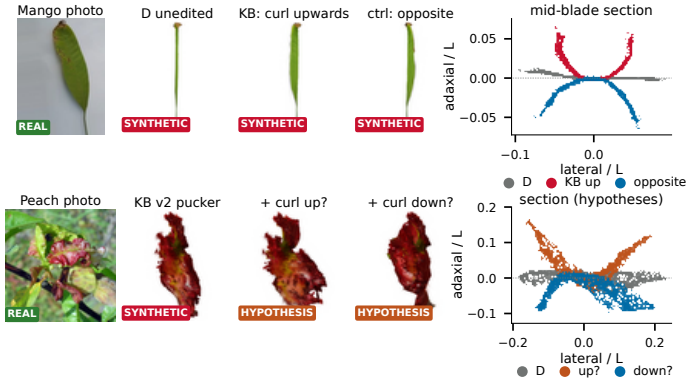
KB colour recall 0.71 (C) and 0.69 (C-stage) vs 0.38 generic; wrong-stage and other-disease controls fall to 0.25 / 0.29. So KB colour gains are *not* evidence for 3D.

2D can change shape, but not correctly

Arm I (InstructPix2Pix) changes the outline (0.83) but flattens the real puckering. A KB-constrained 2D warp baseline is pending.

REAL SAGE DATA | OUR RUN – MEASURED Peach 0087 at the input view: real photo and 2D arms. results/statistics/metrics2d_summary.json (n per arm on the chart). Colour detectors: Lab shifts, $T = 5$. "Purple" detector and "pale" definition are POST-HOC CHANGE (LOGGED); both versions in report 05.

Test 1: Directional Shape From the Text



Positive control: the operator follows a direction

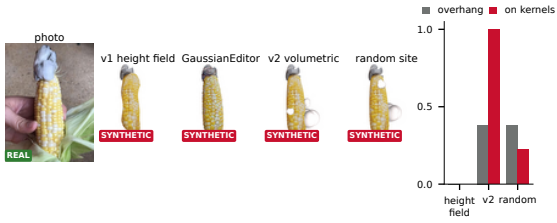
Counterfactual (leaves have anthracnose): "curl upwards" raises margins 3/3 (4.8° →46.2°), reversed 3/3 (-35.4°). IoU 0.91→0.62: contradicts the photo. Verifies code, not anatomy.

Peach: the source gives no direction

The operator **abstains**; up/down only as labelled hypotheses. Pucker on reddish tissue: enrichment 1.14 vs 1.07 uniform.

SYNTHETIC – UNVERIFIED **OUR RUN – MEASURED** results/demos/{mango_curl,peach}/*/metrics.json; roll = signed margin angle about the midrib (+ = toward the upper side). Recovering an imposed angle is a compliance check; a real curl phenotype is still needed.

Test 2: Galls as Volumes on Kernel Tissue



What a height field cannot do

v1 only pushes the surface out (no overhang). v2 galls are closed lobed bodies; 0.38 of their surface overhangs the ear.

Placement from evidence

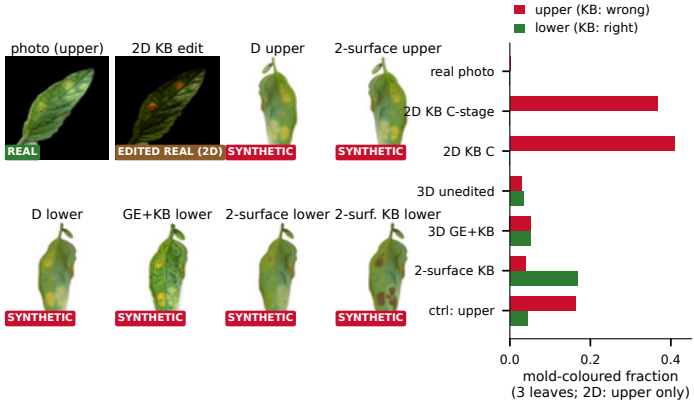
"...develop on the ear replacing kernels" (UNL): on kernel tissue in 1.00 of runs vs 0.22 random: *constraint compliance*, not a finding.

What is assumed

Ear length 18 cm (no ruler); count and position are in no source. 1 ear abstained (no kernel tissue).

REAL SAGE DATA **SYNTHETIC – UNVERIFIED** **OUR RUN – MEASURED** results/demos/corn/*/metrics.json, n = 2 ears × 3 seeds. Added volume 6.1 vs 4.5‰ of (ear length)^3 (galls vs height field; scale-free); 35.7 vs 26.4 cm^3 only under the assumed 18 cm ear. Not validated against real galls.

Test 3: Upper vs Lower Leaf Surface



2D puts the mold on the wrong side

KB: mold “on the lower leaf surface below leaf spots”. A photo shows the upper side only; KB 2D edits paint mold colour on 17–49% (C-stage) / 31–53% (C) of it; real: 0–0%.

Two surfaces keep it apart

One sheet mirrors upper spots below (12–20% of lower pixels). Two surfaces: mold 7–29% of lower pixels, upper 0–8% (baseline 0–7%); 73–84% of it under real spots (random 7–23%).

REAL SAGE DATA | SYNTHETIC – UNVERIFIED | OUR RUN – MEASURED

results/demos/tomato/*/metrics.json, 3 leaves × 3 seeds, all 18 rig views. Colour classes fixed before the edits. The real underside is *unknown*: this shows the KB statement can be represented, not that it is right.

When the KB and the Photo Disagree: an Explicit Policy

Edit (preserve mode)	accepted	decision
Peach pucker (KB placement)	18/18	✓ keep
Tomato two-surface mold	27/27	✓ keep
Mango curl, cited and reversed	refused 18/18	counterfactual
Peach curl up / down	–	abstain
Corn galls on kernels (Test 2)	refused 4/6	see right

Rule. *Preserve*: refuse an edit whose fitted input-view IoU drops by more than $\tau = 0.046$ (the largest drop of appearance-only edits). *Counterfactual*: allowed, always labelled. *Abstain*: the KB is silent.

The photo constrains the galls

Hidden-side centres are not enough (IoU 0.94→0.77): KB-sized galls stick out past the silhouette. Render-checked placement (sample, render the input view, keep if within τ) succeeds in 6/6 runs (1–8 tries).

. . . and forces their size

Corn 0012: gall diameter 0.061 vs 0.094 of ear length unconstrained; volume 0.6 vs 6.1‰. Real galls in the photos: 0.09–0.14 (size plausibility only).

OUR RUN – MEASURED evidence_policy.py; results/review b2 and b3. τ comes from other arms but was set after the pilot existed; decisions at 2τ in the report.

Is It Really 3D? Fair 2D Baselines and All 12 Leaves

Direction: best-case 2D warp

The exact projection of the 3D curl into the photo barely separates up from down (ΔE 5–14); the 3D assets differ by 41–46 over 18 views. A single image hardly encodes direction.

Surface: two-layer 2D

Front image + edited back image is as good in one-surface views. It must be wrong on the minority surface: 0–23% of leaf pixels (Tomato 7–13%, 5–8 of 18 views mixed). 3D helps at oblique views and on non-flat leaves.

Curl on every leaf

Direction follows the command on 11/11 leaves (up and reversed), but magnitude varies 7–43° for one target; midrib kept (length $\geq$0.985). Fails on 2 Citrus leaves (strain p95 1.2–1.7 vs 0.07–0.41).

Reading. 3D adds direction and surface identity that 2D cannot hold, but only partly for flat leaves, and the curl operator does not transfer to every leaf shape. Every curl contradicts its photo (0/22 pass preserve mode; IoU drop 0.07–0.38): a real curl phenotype is still needed.

OUR RUN – MEASURED results/review b1_curl (11 leaves + ring 0006 excluded), b4_surfaces, b7_warp2d. Strain, midrib and roll profile are independent of the imposed angle.

Answer: What Does KB-Guided 3D Add Beyond 2D?

Capability	Measured here	Status
Which *surface* a symptom is on	2D KB edits: mold on the upper side (17–49%); two-layer 2D wrong on 0–23% of pixels	● partly beyond 2D
Which *direction* a leaf deforms	follows cited direction 3/3 (counterfactual Mango); Peach abstains	● compliance only
Symptom *volume* and overhang	closed galls on kernel tissue (1.00); height field: no overhang	✓ representable
One object across views	3D 2.0–10.4 vs per-view 2D 16.8–19.3; 2D painted on it 2.7	● from sharing
Hidden symptoms are *correct*	no suitable real views in datasets examined	NOT RUN / BLOCKED
Biologically realistic	no ratings yet (package ready)	NOT RUN / BLOCKED
Better downstream classifier	minimal test: NOT SUPPORTED (falsified at this scale)	✗ not supported

Direct answer. KB-guided 3D can *express* the direction, volume and (partly) the surface of a symptom, which one image cannot. Photo agreement is *measured*, not guaranteed: kept for underside and photo-checked galls, broken by every curl. Whether the shapes are *true* or *useful* is not shown yet.

OUR RUN – MEASURED NOT RUN / BLOCKED results/demos/summary.json; results/downstream_minimal/summary.json; reports/11–14 in the SAGE-Extend repository.

Ablations: Knowledge × Geometry

2×2 on total KB recall

	no KB	KB
no geometry	D 0.00	K 0.17
geometry	G 0.04	H 0.53

Interaction $(H-G)-(K-D) = +0.32$, **n = 6** specimens with every cell (2 classes).

Ablation	Arms	Status
No KB / generic	B, E, G	✓
KB, wrong stage	C-NG	✓
Mismatched KB (neg. ctrl)	C-MM	✓
KB appearance only	K	✓
KB geometry only	J	✓
Appearance vs geometry 3D	K vs J vs H	✓
Placement: evidence vs random	J2/J2u, V/Vr, BLk/BLr	✓
Direction: cited vs opposite	CU/CD	✓
Viewpoints (1/2/4/8/adaptive)	–	✗
Severity, occlusion, background	–	✗

OUR RUN – MEASURED results/statistics/ablation_2x2.json. Total KB recall = mean of geometric and colour recall over the stage's required attributes.

Statistics: Specimen Is the Unit

Comparison	n (classes)	means	Δ [95% CI]	Holm p	Reading
KB geometric recall, J vs G	6 (2)	0.44 / 0.07	+0.37 [+0.13, +0.61]	0.88	direction only, 2 classes
Unsupported attributes, J vs G	9 (3)	0.06 / 0.17	[-0.28, +0.03]	1.00	inconclusive
Coherence, J vs E	9 (3)	0.81 / 0.31	[+0.37, +0.61]	0.043	survives Holm
Texture grain, J vs E	9 (3)	1.28 / 12.94	[-15.89, -7.94]	0.043	survives Holm
Texture grain, K vs E	15 (5)	8.71 / 16.98	[-14.15, -4.24]	0.001	survives Holm
Texture grain, H vs F	9 (3)	7.09 / 15.73	[-11.96, -5.45]	0.043	survives Holm

Procedure

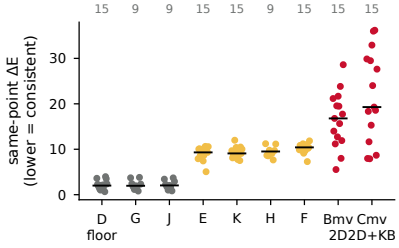
Seeds averaged within specimen; paired Wilcoxon; Holm correction across comparisons; specimen bootstrap CI and class-cluster bootstrap (flagged when <3 classes).

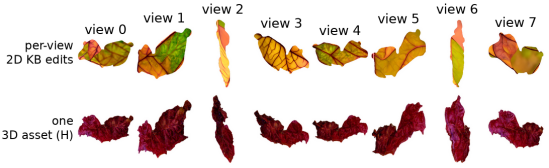
What this design can detect

KB recall has 4 non-tied pairs ($n = 6$): the exact test cannot go below $p = 0.125$, Holm over 13 tests ≥ 0.875, at *any* effect size. Only artefact measures (9–15 specimens) can survive, and do. More disease classes, not more seeds.

OUR RUN – MEASURED results/statistics/comparisons.json; reports/07; results/review/b5_statistics (paired per-specimen values, leave-one-class-out). A bootstrap CI over 6 specimens can exclude 0 while no exact test can: the CI is anti-conservative at this *n*.

E4: One 3D Asset vs Per-View 2D Edits





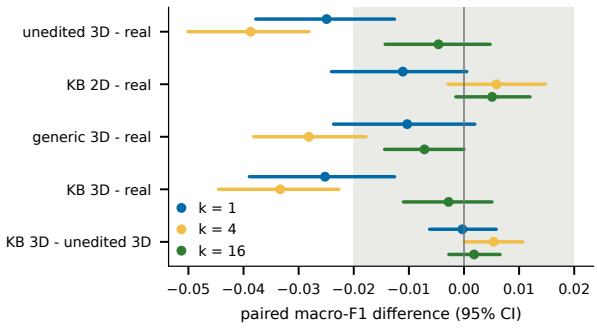
Per-view 2D: 16.8 (generic), 19.3 (KB). 3D GaussianEditor: 9.3–10.4. Geometry editor: 2.0, the floor 2.0.

Metric. Same surface point, adjacent views (depth-tested): colour difference ΔE; unedited D = floor. **Partly free:** one shared asset is rendered for every view, so any 3D representation gains some consistency, and doing nothing scores the floor.

Read with edit size. Colour change vs D: GaussianEditor 20.2/18.4 (E/F), geometry editor 4.9 (it moves shape: silhouette change 0.14, not colour); 2D edits 36–47 inside their mask. J sits at the floor because it barely recolours. A 2D KB edit *painted onto the shared asset* is just as consistent (2.7) while recolouring by 32: the gain is the shared asset, not our editor.

SYNTHETIC – UNVERIFIED OUR RUN – MEASURED results/statistics/crossview_corr_*.json; n per arm on the chart (Bmv/Cmv on the 4 Peach/Tomato specimens with masked per-view edits).

E5: Minimal Downstream Test on Real Photos

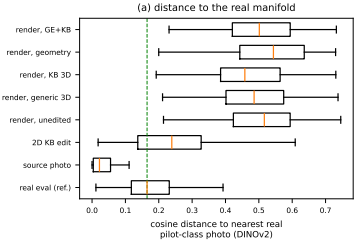


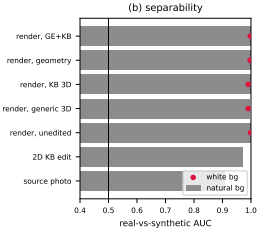
k real/class	KB-3D − real only	KB-3D − unedited 3D
1	-0.025	-0.000
4	-0.033	+0.005
16	-0.003	+0.002

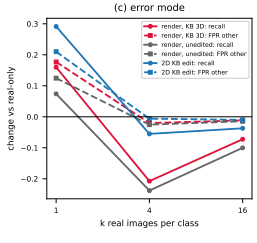
Verdict (pre-registered): NOT SUPPORTED (falsified at this scale). The knowledge edit adds under 2 macro-F1 points over unedited renders; renders themselves hurt at small *k*. The render-to-photo gap comes before knowledge.

OUR RUN – MEASURED reports/14_downstream_minimal.md: within-crop disease classification, 5 crops, frozen DINOv2 + linear probe, 50 paired draws per *k*, 2,936 held-out real SAGE photos, pHash-disjoint. Macro-F1 differences; 95% CIs in results/downstream_minimal/contrasts.csv.

E5: Why Renders Hurt – the Render-to-Real Gap







Off the real manifold

Nearest-real distance: renders 0.52/0.46 (unedited/KB) vs 0.16 real-to-real; 2D edits 0.24.

Not the background

Real vs render AUC 1.00; white background still 0.99: the rendering itself.

Shift, not information

At $k = 4$ pilot recall -20.8 (KB) / -23.8 (unedited) points; other errors barely move.

OUR RUN – MEASURED gap_analysis.py (cached features) and a matched pilot (results/review/b8): render at the photo's pose +0.28, white background +0.09, new views +0.21, photo colours -0.01. Fix rendering and viewpoints, not colour.

Validation: Raters and Real Multi-View Data

Expert review package (built, not yet sent)

84 blinded, shuffled items (72 test, 9 repeats, 3 catch); real photo beside each candidate; verbatim KB text; 6 anchored 1–5 criteria incl. surface and shape. About 57 min per rater. Agreement code (Krippendorff α, weighted κ, ICC) passes 14/14 tests against textbook values.

Limit

2D vs 3D cannot be fully blinded: one photo and a multi-view strip look different.

Matched multi-view data

Among the $\sim$30 public datasets we examined, **none** pairs both leaf surfaces, 3D curl, or gall volume for a diseased specimen (others may exist). Protocol: 30 specimens (8 per disease + controls), 98 views each incl. an underside ring, flat photos of both surfaces, one view as input, the rest held out.

Timing

Corn smut: July–Sept.; Tomato Leaf Mold: greenhouse; Peach Leaf Curl: April–May 2027. About 45 person-hours.

NOT RUN / BLOCKED reports/11_multiview_data_search.md, 12_acquisition_protocol.md, 13_expert_review_protocol.md; results/review_package_v1/ (key file kept separately).

Compute: Two 12 GB TITAN X GPUs

Step	median wall-clock	runs ok / tried
SAM 3D reconstruction (offloaded)	1070 s	15 / 15
Pose alignment A0–A3	78 s	15 / 15
Geometry edit J or G (+18 renders)	44–45 s	54 / 54
IP2P 2D edit, GPU / CPU	57 / 175 s	444 / 444
GaussianEditor E / F, 200 steps	2211 / 2153 s	16/17, 15/20
GaussianEditor K / H (frozen)	2133 / 2159 s	24 / 24

Peak VRAM: SAM 3D 11.6 GB of 12. Counts are *attempts*; the Status slide counts *finished specimens* (E, F: 15/15). The 6 failed attempts (1 E, 5 F) were the two bugs in the repository audit (backup), rerun after the fix; one E success (Peach 0006) was an invalid no-op and was redone.

≈3.4 GPU-h

per specimen for the full arm set

≈50×

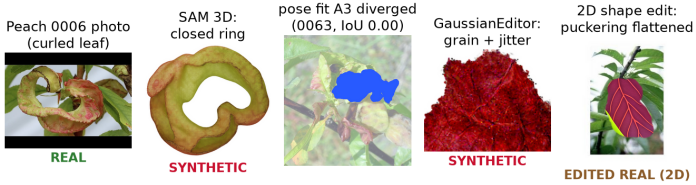
GaussianEditor vs geometry editor run time

(2251 s vs 44 s per edit)

The cheap editor is also the one that produces the shape change: cost is not the bottleneck for scaling J/H.

OUR RUN – MEASURED reports/08_gpu_compute_audit.md, generated from JSONL run logs and nvidia-smi samples (results/compute/gpu_samples.csv).

Failure Taxonomy



Reconstruction

Ring topology; 13/15 over-scaled; one-sheet leaves.

→ ✓ pose fix A3*; ✓ two-surface leaf (Test 3); ● reject closed rings before editing.

Optimisation

Pose refinement diverged on 3/15.

→ ✓ accept-if-better rule; ● add held-out depth as a second gate (worse on 6/15).

Editing

GE: grain, jitter, no shape; J blisters at random.

→ ✓ geometry editor for shape; ✓ evidence placement (weak gain); ● better lesion detector.

Biology

2D edit removes real puckering; stage mixing.

→ ✓ stage-aware KB; ● observed-morphology constraint; expert ratings blocked.

REAL SAGE DATA | SYNTHETIC – UNVERIFIED reports/09_failure_analysis.md; 00_full_failure_audit.md (root cause and status of each). ✓ done, ● next.

Execution Plan: Validate Before Generating More

01 Blinded plant-pathologist ratings
84-item package ready; 2–3 raters; one primary outcome declared first.

02 Small matched multi-view set
Corn smut and Tomato Leaf Mold first, with a ruler; held-out views score surface, direction, volume.

03 Independent geometry evaluation
Reference curl, gall volume and surface localisation, not the operators' own detectors.

04 Render-to-real diagnosis
Small matched pilot (background, lighting, texture, pose) before any 250 GPU-h E5.

05 Stronger baselines
KB-constrained 2D warp, two-layer 2D, shared-texture multi-view 2D.

Blocker	Unblocked by
No raters	3 recruited experts (01)
No matched real views	specimen capture (02)
No physical scale	ruler in captures
Render-to-photo gap	diagnosis (04)
Peach curl direction	a verbatim source

Question for you. Can we fund about 45 person-hours of capture and recruit three raters? Both precede any validity claim.

OUR RUN – MEASURED order follows the external review (validation first). Full blocker list and deliverables: backup slides; reports/12–15.

Go / No-Go After the Pilot

Decision	Required evidence vs what we have	Verdict
GO: KB-grounded 3D reconstruction	Verified hidden-trait recovery and a measured gain over KB 2D. *No held-out real views exist; minimal downstream test: no gain.*	✗ not met
PIVOT: 3D simulation and inspection	Plausible, controllable shape edits with limited claims. *Surface, direction and volume are representable and measured; consistent views; not yet expert-rated.*	● **best fit**
PIVOT: 2D knowledge-guided editing	Knowledge helps, 3D adds nothing. *Knowledge does steer 2D colour; but 2D cannot represent shape correctly (arm I).*	● partly
NO-GO: unreliable synthetic biology	Errors dominate. *Failures are real (ring, GE jitter) but diagnosed, reproducible and avoidable.*	✗ not dominant

Recommendation. Narrow the claim to *knowledge-controlled 3D symptom simulation*: an auditable KB→shape editor whose edits are traceable to cited text and consistent across views. Do not claim reconstruction of hidden anatomy. Numerical go/no-go thresholds will be set from pilot variances, not before.

OUR RUN – MEASURED evidence from slides E1–E4; decision logic from reports/10_final_experimental_protocol.md (Outcome D for now: no established advantage on real data).

Selected References

[1] SAM 3D Team, X. Chen *et al.*, “SAM 3D: 3Dfy Anything in Images,” *arXiv:2511.16624*, 2025.

[2] Y. Chen *et al.*, “GaussianEditor: Swift and Controllable 3D Editing with Gaussian Splatting,” *CVPR*, 2024.

[3] B. Kerbl *et al.*, “3D Gaussian Splatting for Real-Time Radiance Field Rendering,” *ACM TOG*, 2023.

[4] T. Brooks, A. Holynski and A. A. Efros, “InstructPix2Pix: Learning to Follow Image Editing Instructions,” *CVPR*, 2023.

[5] A. Haque *et al.*, “Instruct-NeRF2NeRF: Editing 3D Scenes with Instructions,” *ICCV*, 2023.

[6] O. Sorkine and M. Alexa, “As-Rigid-As-Possible Surface Modeling,” *Symposium on Geometry Processing*, 2007.

[7] Disease text: SAGE KB source pages (Ohioline PLPATH-FRU-26; UMN Extension; APSnet; Plantix) and UNL CropWatch (corn smut), quoted verbatim.

Backup Slides

Backup

Status and repository audit, the twelve questions, blockers, deliverables, GaussianEditor, original prototype.

Backup: Status at a Glance

15

pilot specimens, 5 SAGE classes, chosen before any model was run

15/15

SAM 3D reconstructions, pose-corrected

9/15

specimens with KB and generic geometry edits (×3 seeds)

Arm	specimens done	state
2D KB-stage / shape edits (C-stage, I)	15/15	complete, 3 seeds
GaussianEditor generic (E)	15/15	complete
GaussianEditor + KB (F)	15/15	complete
KB colour, geometry frozen (K)	15/15	complete
Geometry: generic G / KB J	9/15	done where operator exists

Gate respected. Scale-up waited until Phase A (Peach 0087, every arm) showed coherent shape edits. Phase B finished on two 12 GB TITAN X (about 38 min per GaussianEditor run); the v2 tests take seconds per edit.

OUR RUN – MEASURED counts of asset files under results/pilot and results/phaseA at 2026-10-15 04:22; geometry arms exist only where the KB names an implemented operator: Peach (distort, pucker, thicken), Corn (outgrowth), Mango (depress; its detector is excluded as non-specific). Tomato and Citrus have none.

Backup: What the Repository Contained (Phase 0)

As found (verified against files and logs)

Claimed	Found
100–150 examples	✗ **one** mango-leaf screenshot
GaussianEditor edits	✗ never ran (env build failed)
SAM 3D needs an A100	✗ runs on 12 GB TITAN X with stage-wise offload (peak 11.6 GB)
3D adds information	✗ that leaf is near-planar (z-extent 4% of length)

Bugs found and fixed while running

01 **KB arm F never ran**
The colon in every KB prompt was parsed as a config key. Prompts now go through validated config files; the prompt the model receives is checked (10 unit tests).

02 **“No visible change” on Peach 0006**
SAM 3D writes infinite opacities → NaN loss → every fp16 step skipped. Sanitised before editing.

03 **GPU 1 crashed at start**
An allocator setting inherited from SAM 3D is unsupported by GaussianEditor’s torch 2.0.1.

04 **Per-view 2D baseline unmasked**
IP2P invented new scenes; re-scored with the object mask.

OUR RUN – MEASURED reports/phase1_protocol/00_repository_audit.md (as-found state), reports/00_full_failure_audit.md (each bug: location, cause, evidence, fix, test).

Backup: The Twelve Questions (1/2)

#	Question	Answer from the evidence
1	Why is 3D necessary here?	To state which surface, which direction and how much volume (Tests 1–3). For flat spots it adds nothing.
2	Which symptoms gain from 3D?	Underside sporulation (Tomato Leaf Mold, 67 SAGE records name a lower surface), galls (Corn Smut), directional curl (42 records name curl/roll; direction is given in some, not for Peach).
3	How reliable is one-image reconstruction?	Input view after pose fix: median fitted IoU 0.92 (a fit, not accuracy); 1/15 topology failure. Unseen views: unknown.
4	What does the KB contribute?	Which attributes to edit, at which stage, and which edits are prohibited. It raises KB recall in 2D (0.71 vs 0.38) and 3D shape (0.44 vs 0.07).
5	Can it be independently verified?	Text-level yes (cited phrase per edit). Specimen-level no: needs real multi-view data or expert raters.
6	Hidden symptoms recovered or just plausible?	Represented, not recovered. Nothing hidden can be verified with SAGE or the public sets we examined.

OUR RUN – MEASURED all numbers are macros generated from results/ (2026-10-15 04:22).

Backup: The Twelve Questions (2/2)

#	Question	Answer from the evidence
7	Is 3D editing better than KB 2D editing?	For colour: no measured advantage. For shape: 2D either cannot change it (mask) or changes it wrongly (arm I); 3D J/H can. For consistency: 3D far better (2.0–9.3 vs 16.8–19.3).
8	Knowledge vs geometry share of the gain?	2×2 on 6 specimens: interaction +0.32; each factor adds a different attribute type.
9	Does it improve a real agricultural task?	Minimal test on real photos: not supported (KB-3D vs unedited renders $\leq$ +0.02 macro-F1). Full E5 not justified yet.
10	Strongest failure modes?	Topology failure on curled leaves; GaussianEditor jitter and grain; render-to-photo gap; edits that contradict a flat photo (curl lowers input IoU); no ruler for physical size.
11	Experiments still required?	Send the expert package; capture 30 multi-view specimens (protocol ready); close the render gap; then E5.
12	Ready for publication?	Not yet as "reconstruction". A narrower paper (auditable KB$\rightarrow$3D symptom editing, with honest limits) is within reach after expert review.

OUR RUN – MEASURED **NOT RUN / BLOCKED** item 9: reports/14; item 11 depends on data or raters we do not yet have.

Backup: Blockers in Full

Blocker	Effect	Unblocked by
No matched real views	hidden surface, direction and volume cannot be scored	capture protocol (30 specimens)
No expert raters	biological realism unmeasured	package v1, 3 raters
No physical scale in photos	gall size in cm rests on an assumed ear length	ruler in future captures
Peach curl direction not in source	curl abstains for the demo disease	a verbatim source
Render-to-photo gap	synthetic data hurts a real classifier	rendering work before E5
Single-image reconstruction errors	ring topology (Peach 0006), over-scale	multi-view input
Disk on the workstation	13 GB free	archive old runs

NOT RUN / BLOCKED full list with owners: reports/15_revised_scientific_report.md, section “Blockers”.

Backup: Deliverables Checklist

Deliverable	built	tested	real data	Blocked by / evidence still needed
Code audit; failure audit	✓	✓	✓	–
Ranked case shortlist	✓	✓	✓	expert check of SAGE labels
Dataset and ground-truth protocol	✓	✓	●	Tier 1/2 data
Structured KB + validator	✓	✓	✓	expert check of operator mapping
Baseline arms A–K, shared runner	✓	✓	✓	–
E1 reconstruction	✓	✓	●	held-out real views
E2 editing fidelity	✓	✓	●	expert ratings
E3 three focused tests (v2 editor)	✓	✓	●	held-out real views
E4 multi-view consistency	✓	✓	●	(accuracy needs real views)
E5 minimal downstream	✓	✓	✓	full study gated
Expert review package	✓	✓	–	raters
Multi-view search + protocol	✓	–	✗	capture
Ablation matrix	✓	●	●	viewpoint, severity ablations
Statistics plan and code	✓	✓	✓	larger *n*
QC, compute logging, resume	✓	✓	✓	–
Figure code from results	✓	✓	✓	–
Revised report (reports/15)	✓	–	–	–

OUR RUN – MEASURED reports/00–10, NEXT_EXPERIMENT_RESULTS.md, RESEARCH_PROGRESS.md in the SAGE-Extend repository. ✓ done ● partial ✗ not done.

Backup: GaussianEditor (CVPR 2024)



Teaser (Fig. 1) from Chen *et al.*, CVPR 2024; github.com/buaacyw/GaussianEditor.

Semantic tracing. Edits only the target Gaussians.

Hierarchical splatting. Stable under 2D guidance.

Diffusion guidance. InstructPix2Pix edits rendered views from text; Gaussians are fit to them.

Our finding. On SAM 3D leaves it changes colour, not shape (E2).

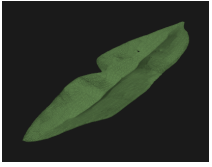
Backup: The Original One-Leaf Prototype



Rule-based lesions



Yellowing, holes



SAM 3D mesh

Where we started

One mango-leaf screenshot, rule-based re-colouring, GaussianEditor not yet running. The leaf is nearly planar, and anthracnose is a surface symptom: a poor test of what 3D adds. This is why Phase 1 re-selected the cases.

Earlier deck (October 2026), figures from github.com/ushashi149/SAGE-Extend (commit 371d96b).